

Student Name:

Student ID:

Habib University
Dhanani School of Science and Engineering
Computer Architecture
EE 371 / CS 330 / CE 321 (Fall 2025)

Quiz 4

Date: 1st Dec, 2025

Total marks: 20

Obtained Marks:

Total time: 45 mins

Instructions:

1. All questions should be answered **with ink pen**.
2. Smart watches, laptops, and similar electronics are **strictly NOT allowed**.
3. Answer sheets should contain all steps, working, explanations, and assumptions.
4. Write your name and HU ID on all sheets.
5. Turn in your question paper along with your answer sheets.
6. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This quiz assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	✓

Student Name:

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- In this question, we will design a cache and find out first word address for each of the cache index. Suppose we have 512 bytes SRAM. We want to arrange it in 4 words per block direct mapped cache. Draw a table to showing first three and last cache indexes, first block address/number and first word address that will could be stored there for each index. **(6 points)**

Sol:

Cache index	Block address/number	Word address
0	0	0
1	1	16
2	2	32
...	...	...
...	...	...
29	29	464
30	30	480
31	31	496

- Assume we have direct mapped cache having 8 words per block. The total size of this cache is 16KiB and processor has 32-bit addressing.

a) How many bits do we need for tag, index and word offset?

Sol:

$$2^m = 8 \rightarrow m = 3 \text{ bits (word offset)}$$

$$\text{Total blocks} = (16 \times 1024) / (8 \times 4) = 512 \text{ blocks} \rightarrow 2^n = 512 \rightarrow n = 9 \text{ bits (index)}$$

$$\text{Tag} = 32 - n - m - \text{byte_offset} = 32 - 9 - 3 - 2 = 18 \text{ bits}$$

b) Calculate total number of bits needed for the cache.

Sol:

$$\text{Single cache block/line} = \text{Tag bit} + \text{Valid bit} + \text{block size}$$

$$\text{block size} = 8 \times 4 \times 8 = 256 \text{ bits}$$

$$\text{Single cache block} = 18 + 1 + 256 = 275 \text{ bits}$$

$$\text{Total bits} = \text{single cache block} \times \text{total blocks} = 275 \times 512 = 140800 \text{ bits or } 17.18 \text{ KiB}$$

c) For the following addresses, find block address and cache index for the given cache

1400

Sol:

$$\text{Block address} = \text{word address/bytes per block} = 1400 / (8 \times 4) = 43.75 \rightarrow 43$$

Student Name:

Student ID:

$$\text{Index} = \text{Block address} \% \text{No. of cache blocks} = 43 \% 512 = 43$$
$$57600$$

Sol:

$$\text{Block address} = \text{word address/bytes per block} = 57600/(8*4) = 1800$$

$$\text{Index} = \text{Block address} \% \text{No. of cache blocks} = 1800 \% 512 = 264$$
$$172800$$

Sol:

$$\text{Block address} = \text{word address/bytes per block} = 172800/(4*8) = 5400$$

$$\text{Index} = \text{Block address} \% \text{No. of cache blocks} = 5400 \% 512 = 280$$
$$1363968$$

Sol:

$$\text{Block address} = \text{word address/bytes per block} = 1363968/(4*8) = 42624$$

$$\text{Index} = \text{Block address} \% \text{No. of cache blocks} = 42624 \% 512 = 128$$

- d)** If words per block is kept constant and we convert this cache into a 4-way set associative cache, find number of bits for the tag and index part.

Sol:

$$\text{We will have } 1/4 \text{ of the total cache blocks} \rightarrow \text{index} = \text{index}/4 = 512/4 = 128$$

$$2^n = 128 \rightarrow n = 7 \text{ bits (index)}$$

$$\text{Tag} = 32 - n - m - \text{byte_offset} = 32 - 7 - 3 - 2 = 20 \text{ bits}$$

- e)** Assume a processor is running at 4GHz. The main memory has an access time of 200 clock cycles. For this processor, a perfect cache (i.e. no misses) has a base CPI of 1.5. Find the new CPI for this processor if it has a L1 instruction cache with a miss rate of 3% and L1 data cache with a miss rate of 1%. Assume that the data instructions constitute 50% of the total instructions.

Sol:

$$\text{miss penalty} \rightarrow 200 \text{ cc}$$

$$\text{CPI}_{\text{new}} = \text{CPI}_{\text{base}} + \text{CPI}_{\text{cache}}$$

$$\text{CPI}_{\text{cache}} = 3\%*200 + 1\%*200*50\% = 6 + 1 = 7$$

$$\text{CPI}_{\text{new}} = 1.5 + 7 = 8.5$$